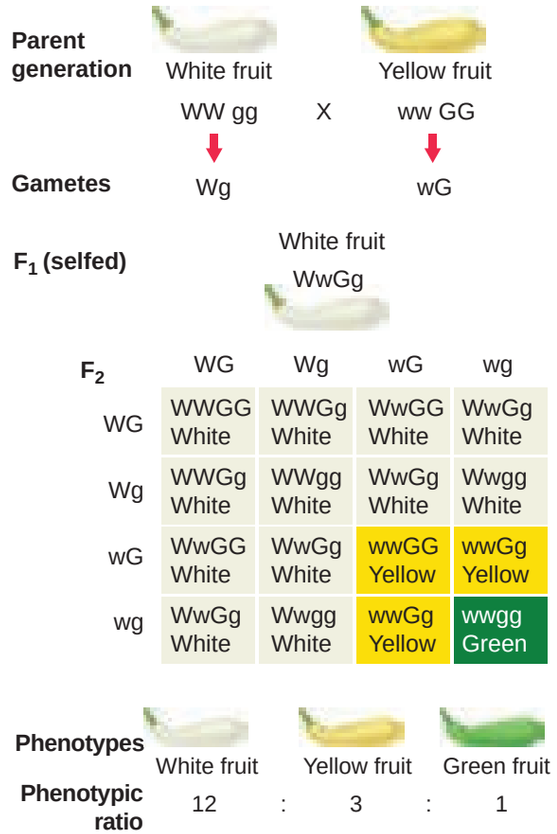


Figure 2.16: Dominant epistasis in summer squash

Since W is epistatic to the alleles 'G' and 'g', the white which is dominant, masks the effect of yellow or green. Homozygous recessive ww genotypes only can give the coloured fruits (4/16). Double recessive 'wwgg' will give green fruit (1/16). The Plants having only 'G' in its genotype (wwGg or wwGG) will give the yellow fruit(3/16).

Intra –genic or allelic interaction

S. No.	Gene interaction	Example	F_2 Phenotypic ratio
1	Incomplete Dominance	Flower colour in *Mirabilis jalapa.* Flower colour in snapdragon (*Antirrhinum* spp.)	1 : 2 : 1 1 : 2 : 1
2	Codominance	ABO Blood group system in humans	1 : 2 : 1

Table 2.2: Intra- genic interaction

Inter-genic or non-allelic interaction

S. No.	Epistatic interaction	Example	F_2 Ratio Phenotypic ratio
1	Dominant epistasis	Fruit colour in summer squash	12 : 3 : 1
2	Recessive epistasis	Flower colour of *Antirrhinum* spp.	9 : 3 : 4
3	Duplicate genes with cumulative effect	Fruit shape in summer squash	9 : 6 : 1
4	Complementary genes	Flower colour in sweet peas	9 : 7
5	Supplementary genes	Grain colour in Maize	9 : 3 : 4
6	Inhibitor genes	Leaf colour in rice plants	13 : 3
7	Duplicate genes	Seed capsule shape (fruit shape) in shepherd's purse Bursa bursa-pastoris	15 : 1

Table 2.3: Inter-genic interaction

2.6 Polygenic Inheritance in Wheat (Kernel colour)

Polygenic inheritance - Several genes combine to affect a single trait.

A group of genes that together determine (contribute) a characteristic of an organism is called polygenic inheritance. It gives explanations to the inheritance of continuous traits which are compatible with Mendel's Law.

The first experiment on polygenic inheritance was demonstrated by Swedish Geneticist H. Nilsson - Ehle (1909) in wheat kernels. Kernel colour is controlled by two genes each with two alleles, one with red kernel colour was dominant to white. He crossed the two pure breeding wheat varieties dark red and a white. Dark red genotypes $R_1R_1R_2R_2$ and white genotypes are $r_1r_1r_2r_2$. In the F_1 generation medium red were obtained with the genotype $R_1r_1R_2r_2$. F_1 wheat plant produces four types of gametes R_1R_2, R_1r_2, r_1R_2, r_1r_2. The intensity of

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